Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3341-WE01

Title: (BEng/MEng)

Environmental Engineering 3

Time Allowed:	2 hours
Additional Material provided:	Yes - Formula Sheet included as part of exam paper.
Materials Permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	This is an in-person, invigilated exam. Answer ALL questions. Write your answers in the white-covered answer booklet with barcodes. Begin your answer to each question on a new page. All relevant workings must be shown.

Revision:	

Q1

(a) i. Balance the equation: $NH_4^+ + O_2 \longrightarrow NO_2^- + H_2O + H^+$ [**10%**]

ii. Balance the equation: $NO_2^- + O_2 \longrightarrow NO_3^-$ [**10%**]

iii. Explain where the above two reactions play a role in biogeochemical cycling and why they are represented as two separate equations in the environment. [**30%**]

(b) In your own words, give at least two examples of anthropogenic disruption to the nitrogen cycle. In your answer, describe the disruption and consequences to planetary and/or human health where appropriate. [**50%**]

Q2

(a) i. The organic content of a wastewater is given in Table 1. Write the empirical formula of the organic material such that it is standardised to 1 mole of organic nitrogen. **[35%]**

ii. Determine the chemical oxygen demand (COD) of a wastewater sample containing 32 g L^{-1} of butyrate. Note that one mole of butyrate is 88.11 g and the butyrate reduction half-reaction is provided in the formula sheet. **[15%]**

Table 1: Organic content of a wastewater.

C	114 mg L^{-1}
H	20 mg L^{-1}
O	42 mg L^{-1}
N	24 mg L^{-1}

(b) i. A farmer operates a continuous stirred tank reactor treating 10 m^3 day^{-1} of wastewater with a chemical oxygen demand (COD) concentration of 24,000 mg L^{-1}. They are achieving 40% removal of COD. Assuming 30% of the COD is transformed to CH_4, determine the rate of CH_4 produced in kg day^{-1}. **[35%]**

ii. In theory, 1 g of chemical oxygen demand (COD) should yield 350 mL of CH_4. Explain why real-world anaerobic digestion facilities generally produce lower methane yields. **[15%]**

Q3

Water flows at a rate, Q, over a sharp crested weir with a rectangular cross-section of breadth, B. The water surface is located at an elevation, h, above the crest of the weir. The following standard assumptions apply:

- Absolute pressure throughout the nappe is at atmospheric pressure.
- The upstream water surface is horizontal right up to the edge of the weir.
- Fluid pressure upstream of the weir is hydrostatic.
- Kinetic energy of the water upstream of the weir is negligibly small compared to pressure energy and gravitational potential energy.

(a) Determine the water depth required to achieve a flow rate of 5600 m^3 $hour^{-1}$ when the breadth of the weir is 1.2 m. [**40%**]

(b) Use the aforementioned assumptions, in conjunction with the Bernoulli equation, to derive an equation for the fluid velocity above the weir as a function of elevation above the crest of the weir. [**60%**]

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Formula sheet

Polluted Environments

One electron equivalent of an organic donor has a chemical oxygen demand (COD) of 8 g.

One mole of CH_4 contains 8 e^- eq.

Butyrate reduction half-reaction:

$$\frac{3}{20}CO_2 + \frac{1}{2}HCO_3{}^- + H^+ + e^- = \frac{1}{20}CH_3CH_2CH_2COO^- + \frac{7}{20}H_2O$$

Propionate reduction half-reaction:

$$\frac{1}{7}CO_2 + \frac{1}{14}HCO_3{}^- + H^+ + e^- = \frac{1}{14}CH_3CH_2COO^- + \frac{5}{14}H_2O$$

Periodic Table of the Elements
1 IA 1A
2 IIA 2A
3 IIIB 3B
4 IVB 4B
5 VB 5B
6 VIB 6B
7 VIIB 7B
8
9 VIII 8
10
11 IB 1B
12 IIB 2B
13 IIIA 3A
14 IVA 4A
15 VA 5A
16 VIA 6A
17 VIIA 7A
18 VIIIA 8A
1 H Hydrogen 1.008
2 He Helium 4.003
3 Li Lithium 6.941
4 Be Beryllium 9.012
5 B Boron 10.811
6 C Carbon 12.011
7 N Nitrogen 14.007
8 O Oxygen 15.999
9 F Fluorine 18.998
10 Ne Neon 20.180
11 Na Sodium 22.990
12 Mg Magnesium 24.305
13 Al Aluminum 26.982
14 Si Silicon 28.086
15 P Phosphorus 30.974
16 S Sulfur 32.066
17 Cl Chlorine 35.453
18 Ar Argon 39.948
19 K Potassium 39.098
20 Ca Calcium 40.078
21 Sc Scandium 44.956
22 Ti Titanium 47.867
23 V Vanadium 50.942
24 Cr Chromium 51.996
25 Mn Manganese 54.938
26 Fe Iron 55.845
27 Co Cobalt 58.933
28 Ni Nickel 58.693
29 Cu Copper 63.546
30 Zn Zinc 65.38
31 Ga Gallium 69.723
32 Ge Germanium 72.631
33 As Arsenic 74.922
34 Se Selenium 78.971
35 Br Bromine 79.904
36 Kr Krypton 83.798
37 Rb Rubidium 85.468
38 Sr Strontium 87.62
39 Y Yttrium 88.906
40 Zr Zirconium 91.224
41 Nb Niobium 92.906
42 Mo Molybdenum 95.95
43 Tc Technetium 98.907
44 Ru Ruthenium 101.07
45 Rh Rhodium 102.906
46 Pd Palladium 106.42
47 Ag Silver 107.868
48 Cd Cadmium 112.414
49 In Indium 114.818
50 Sn Tin 118.711
51 Sb Antimony 121.760
52 Te Tellurium 127.6
53 I Iodine 126.904
54 Xe Xenon 131.294
55 Cs Cesium 132.905
56 Ba Barium 137.328
57-71
72 Hf Hafnium 178.49
73 Ta Tantalum 180.948
74 W Tungsten 183.84
75 Re Rhenium 186.207
76 Os Osmium 190.23
77 Ir Iridium 192.217
78 Pt Platinum 195.085
79 Au Gold 196.967
80 Hg Mercury 200.592
81 Tl Thallium 204.383
82 Pb Lead 207.2
83 Bi Bismuth 208.980
84 Po Polonium [208.982]
85 At Astatine 209.987
86 Rn Radon 222.018
87 Fr Francium 223.020
88 Ra Radium 226.025
89-103
104 Rf Rutherfordium [261]
105 Db Dubnium [262]
106 Sg Seaborgium [266]
107 Bh Bohrium [264]
108 Hs Hassium [269]
109 Mt Meitnerium [278]
110 Ds Darmstadtium [281]
111 Rg Roentgenium [280]
112 Cn Copernicium [285]
113 Nh Nihonium [286]
114 Fl Flerovium [289]
115 Mc Moscovium [289]
116 Lv Livermorium [293]
117 Ts Tennessine [294]
118 Og Oganesson [294]
Lanthanide Series
57 La Lanthanum 138.905
58 Ce Cerium 140.116
59 Pr Praseodymium 140.908
60 Nd Neodymium 144.243
61 Pm Promethium 144.913
62 Sm Samarium 150.36
63 Eu Europium 151.964
64 Gd Gadolinium 157.25
65 Tb Terbium 158.925
66 Dy Dysprosium 162.500
67 Ho Holmium 164.930
68 Er Erbium 167.259
69 Tm Thulium 168.934
70 Yb Ytterbium 173.055
71 Lu Lutetium 174.967
Actinide Series
89 Ac Actinium 227.028
90 Th Thorium 232.038
91 Pa Protactinium 231.036
92 U Uranium 238.029
93 Np Neptunium 237.048
94 Pu Plutonium 244.064
95 Am Americium 243.061
96 Cm Curium 247.070
97 Bk Berkelium 247.070
98 Cf Californium 251.080
99 Es Einsteinium [254]
100 Fm Fermium 257.095
101 Md Mendelevium 258.1
102 No Nobelium 259.101
103 Lr Lawrencium [262]
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continued

Civil Engineering Hydraulics

Flow equation for a sharp crested weir with a rectangular cross-section:

$$Q = \frac{2B}{3}\sqrt{2gh^3}$$

The continuity equation:

$$A_1 v_1 = A_2 v_2$$

The momentum equation:

$$F = \rho Q(v_2 - v_1)$$

The Bernoulli equation:

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

Notation:

A	Cross-sectional area [L^2].
B	Channel breadth [L].
F	Net force in flow direction [MLT^{-2}].
g	Gravitational acceleration [LT^{-2}].
h	Water surface elevation above the crest of a weir [L].
p	Fluid pressure [$ML^{-1}T^{-2}$].
Q	Flow rate [L^3T^{-1}].
v	Fluid velocity [LT^{-1}].
z	Elevation [L].
ρ	Fluid density [ML^{-3}].

END